



MA 530: Graph Theory

Course Code	MA 530
Course Name	Graph Theory
L-T-P-C	L-T-P-C:(3-1-0-4)
Prerequisites	Basic understanding of mathematics
Intended for	UG/PG
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Basics:** Graphs, subgraphs, isomorphism, representation of graphs, degrees, walks, trails, paths, cycles, bipartite graphs. [5 Hours]
- **Trees and connectivity:** Characterizations of trees, minimum-spanning-trees, number of trees, Cayley's formula, shortest path algorithms. [5 Hours]
- **Eulerian and Hamiltonian graphs:** Characterizations, Necessary/sufficient conditions. [4 Hours]
- **Graph Coloring:** vertex coloring, chromatic polynomials, edge coloring. [4 Hours]
- **Planar graphs:** Properties, Euler's formula and its consequences, Kuratowski's Characterization. [6 Hours]
- **Matching and Factorizations:** matching in bipartite graphs, maximum matching in general graphs, Hall's marriage theorem, factorization; Tutte's perfect matching theorem and consequences. [7 Hours]
- **Networks:** The Max-flow min-cut theorem, connectivity and edge connectivity, Menger's theorem. [6 Hours]
- **Graph and Matrices:** Adjacency matrix, Laplacian matrix, Eigen Values. [5 Hours]

Text Books

1. J. A. Bondy and U.S.R Murthy, **Graph Theory with Applications**, Macmillan, 1976.
2. D. B. West, **Introduction to Graph Theory**, Vol:2, Prentice hall, 2001.

References

1. F. Harary, **Graph Theory**, Addison-Wesley publishing company, 1969.
2. R. Diestel, **Graph Theory**, 3rd ed. Graduate texts in mathematics 173, 2005.
3. R. B. Bapat, **Graphs and Matrices**, Vol. 27. London: Springer, 2010.